CLASSIFICATION OF HAND PRESHAPING IN PERSONS WITH STROKE USING LINEAR DISCRIMINANT ANALYSIS

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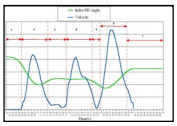
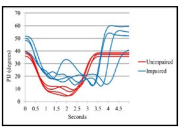
Purpose: Can we measure improvement in the brain's ability to control a multiple degree of freedom hand subsequent to motor training?

Background

- Motor Learning – neural reorganization post rehabilitation supports improved functioning of motor functioning
- Motor Adaptation – modification of movement as a result of repetition across multiple trials
- During motor adaptation the brain modifies its movement across trials to minimize the amount of energy, force, and inaccuracy of movement
- Previous studies employed reach to grasp experiments to measure finger joint angles of subjects with Parkinson's disease and stroke compared to normal subjects, with results indicating significant differences in hand shaping abilities



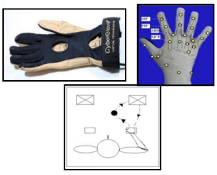


Methods

A. Motor Training

- 4 Subjects trained with interactive video games (VR piano session, Hummingbird game, Hammer game)
- 4 Subjects trained their hand with non-automated repetitive task practice (RTP)

B. Data Capture



CyberGlove® (top left), sensors on CyberGlove® represent hand joints (top right), Schematic of reach to grasp test on bottom

Huge Circle Small Cube

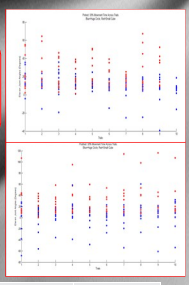
- Flexion and extension of finger joints were measured using resistive bend sensors in a glove (CyberGlove®, Immersion Inc.)
- Hand trajectories were recorded using an electromagnetic sensors (trackSTAR™ system)

C. Data Classification

$$Dij^2 = (xi-\mu j)' \, S^{-1} (xi-\mu j)$$

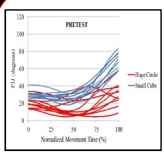
$$P(X|Gj) = f(X|G_j) = \frac{1}{\sqrt{(2\pi)^p \sqrt{|S|}}} e^{(-1/2)D_{ij}^2}$$

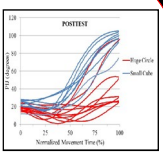
- Eleven dimensional data sample at one time point is classified as one of two shapes
- An observation is classified into one of two classes that has the minimum distance or maximum posterior

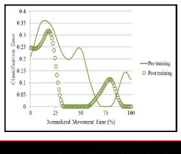


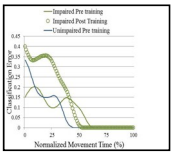
Posteriors	Trial 1	Trial 1	Trial 2	Trial 2
pretest Huge Circle	0.004829	0.995171	3.73E-05	0.999963
pretest Small Cube	3.77E-23	1	1.43E-11	1
posttest Huge Circle	1	4.68E-30	1	5.82E-42
posttest Small Cube	1.82E-39	1	7.58E-99	1

Results









Minimum Classification Error of Hemiparetic Hand: Normalized Movement Time(%)		
	Pre-Training	Post-Training
SUBJECT 1	74%	58%
SUBJECT 2	100%	100%
SUBJECT 3	100%	100%
SUBJECT 4	71%	33%
SUBJECT 5	72%	47%
SUBJECT 6	64%	55%
SUBJECT 7	100%	100%
SUBJECT 8	100%	71%

Conclusion

- Improvement in classification errors or the decrease in error during post training sessions may indicate improvement in an ability to coordinate finger motion.
- Since the classification error results consider eleven finger joint angles from the index to pinky fingers, this analysis may provide insight into improvement in coordination of multiple DOF to pre-shape the hand to the object shape during the reaching movement.

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Bibliography

[1] M. Santello and J.F. Soechting,"Gradual molding of the hand to object contours." *J Neurophysiol,* vol. *79*(3), pp. 1307-1320, 1998
[2] L.F. Schettino, V. Rajaraman, D. Jack, S.V. Adamovich, J. Sage, and H. Poizner, "Deficits in the evolution of hand preshaping in Parkinson's disease," *Neuropsychologia,* vol. *42*(1), pp.82-94, 2000.
[3] P. Raghavan, M. Santello, A.M. Gordon, and J.W. Krakauer,"Compensatory Motor Control After Stroke: An alternative Joint Strategy for Object-Dependent Shaping of Hand Posture," *J Neurophysiol,* vol.*103*, pp.3034-3043, 2010
[4] A.S. Merians *et al.,* "Robotically facilitated virtual rehabilitation of arm transport integrated with finger movement in persons with hemiparesis." *J Neuroeng Rehabil,* vol.8(27), 2011.
[5] A. Shumway-Cook and M.H. Woollacott, *Motor Control: Translating Research into Clinical Practice, Third Edition.* Philadelphia, Pennsylvania: Lippincott Williams & Wilkins, 2007. ch. 18.